

Praditya Wicaksono

AI/ML Engineer & Full-Stack Systems Developer

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PROFESSIONAL SUMMARY

Computational modeling and systems thinking since 2020 — from numerical physics simulation to production AI engineering. AI/ML Engineer and Full-Stack Developer (GPA 3.96/4.00) with a proven track record of building and deploying end-to-end intelligent systems. Experienced in integrating agentic workflows (MCP, LangChain, LangGraph), scalable RAG applications (pgvector, Elasticsearch), and LoRA fine-tuning pipelines into production-grade environments. Published researcher (Scopus-indexed) focused on scalable microservices using FastAPI, robust MLOps practices, and offline-first web systems.

TECHNICAL SKILLS

AI/ML & Research: PyTorch, LLM Fine-Tuning (LoRA, PEFT), LangChain, LangGraph, LlamaIndex, Hybrid RAG (pgvector, Elasticsearch), Vector DBs (Qdrant, Pinecone), Evals (Ragas, TruLens), XGBoost, SHAP

Backend, Infra & MLOps: Python (FastAPI, Streamlit), Node.js, SQL, PostgreSQL, Docker, Kubernetes, GCP, Firebase, CI/CD (GitHub Actions), DVC, MLOps, NeMo Guardrails (LLM Security)

Frontend & Mobile: Next.js (TypeScript), React, Flutter (Dart/Riverpod), Tailwind CSS, ES Modules (JavaScript)

Languages: Indonesian (Native), English (Professional)

PUBLICATIONS & INTELLECTUAL PROPERTY

- **Stock Market Prediction and Time-Series Forecasting using Multi-Head Attention** — Published in *ICAMIMIA 2025* (Scopus-Indexed).
- **Q-POMDP Reinforcement Learning for Adaptive Pedagogical Policy Scheduling** — Published in *IEEE ISemantic 2026*.
- **Lectura Infrastructure & MAESTRO RL Configurator** — Official Indonesian Software Copyright (*HAKI*).

PROJECTS

MAESTRO — Intelligent Tutoring System

Flutter, Riverpod, Machine Learning, Firestore, Groq/Gemini APIs

Adaptive tutoring platform with reinforcement learning

2024 – Present

- Engineered an adaptive learning engine with a modular architecture that dynamically customizes student learning paths in real-time, improving personalized user sessions using reinforcement learning algorithms.
- Implemented a localized, high-performance RAG system utilizing in-memory indexing and LRU caching, significantly reducing LLM response latency by 45% and mitigating content hallucination.
- Deployed the application to production and secured official software copyright (*HAKI*). Live: maestro-4dcce.web.app

JustExplain — Explainable AI Sentencing Predictor

Python, XGBoost, SHAP, MLflow

Collaborative legal sentencing prediction with explainable AI

2025

- Co-developed an ensemble Machine Learning model (XGBoost & Random Forest) trained on 22k+ legal documents from the Indo-Law corpus, achieving 81% F1-score through advanced text embedding and target encoding pipelines.
- Integrated Explainable AI (XAI) using SHAP to provide transparent, interpretable visual breakdowns of model predictions for end-users.
- Deployed an interactive Streamlit dashboard for real-time case simulation, with MLflow tracking for reproducible experiment management.

LLM Fine-Tuning & MLOps Pipeline

Python, Hugging Face, LoRA, MLflow, DVC, Ragas

Parameter-efficient LLM fine-tuning, automated prompt testing and evaluation

2025

- Fine-tuned LLM backbones (Hugging Face, GLM, LLaMA) via LoRA & instruction tuning, reducing trainable parameters by 95%.
- Built end-to-end MLOps pipelines with prompt testing, automated evaluation (Ragas, TruLens), and Git-based model tracking (DVC).
- Tracked 100+ fine-tuning and prompt engineering runs using MLflow to optimize model performance.

Lectura — Offline-First Presentation Engine

Node.js, SQLite, Reveal.js, Docker, Fly.io

High-performance Single Page Application with offline database synchronization

2025 – 2026

- Developed a lightweight, 3-mode Single Page Application (SPA) utilizing a zero-dependency native Node.js server to minimize memory footprint.
- Designed a custom offline-first database synchronization system using SQLite, ensuring robust data persistence.
- Live: lectura-edu-demo.fly.dev

Scalable FastAPI AI Service & Gateway

Python, FastAPI, pgvector, Elasticsearch, Docker, Kubernetes, NeMo Guardrails

High-performance agentic inference service, hybrid RAG database, and secure gateway

2026

- Engineered a scalable FastAPI AI service orchestrated on Kubernetes, ensuring high availability and low latency.
- Designed hybrid RAG retrieval combining pgvector and Elasticsearch for dense/sparse semantic queries.
- Deployed an AI security proxy with LLM guardrails (NeMo Guardrails) to mitigate injection vulnerabilities.

EDUCATION

Universitas Dian Nuswantoro

Semarang, Indonesia

Master of Computer Science (M.Kom.), Informatics Engineering — GPA: 3.96 / 4.00

2024 – 2026

Universitas Negeri Semarang

Semarang, Indonesia

Bachelor of Education (S.Pd.), Physics

2018 – 2023